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Noise-aware multi-channel reconstruction for ultra-low-threshold superfluid helium detectors

Content

The search for sub-GeV dark matter and precision measurements of low-energy neutrino processes motivate detectors capable of stable operation at eV-scale energy thresholds. The DELight experiment employs superfluid helium-4 instrumented with magnetic microcalorimeters (MMC-based LAMCALs), targeting eV-scale baseline resolution and projected nuclear-recoil thresholds near 10 eV. Such performance requires reconstruction methods that remain robust under correlated and time-varying noise in a multi-channel cryogenic read-out.

We present a noise-aware reconstruction framework developed for DELight that combines optimal filtering with correlation-aware analysis across channels. Pulse templates and noise power spectra measured under realistic operating conditions are used to construct statistically consistent estimators, enabling controlled studies of resolution and threshold behavior in the presence of non-stationary noise.

Studies with simulation and calibration R&D data indicate improved stability of near-threshold energy reconstruction and enhanced trigger efficiency compared to purely per-channel approaches. The methods are directly relevant to coherent elastic neutrino–nucleus scattering (CEvNS) and other low-energy neutrino measurements in calorimetric detectors operating close to threshold.

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